Sound and voice processing

Assignment 1

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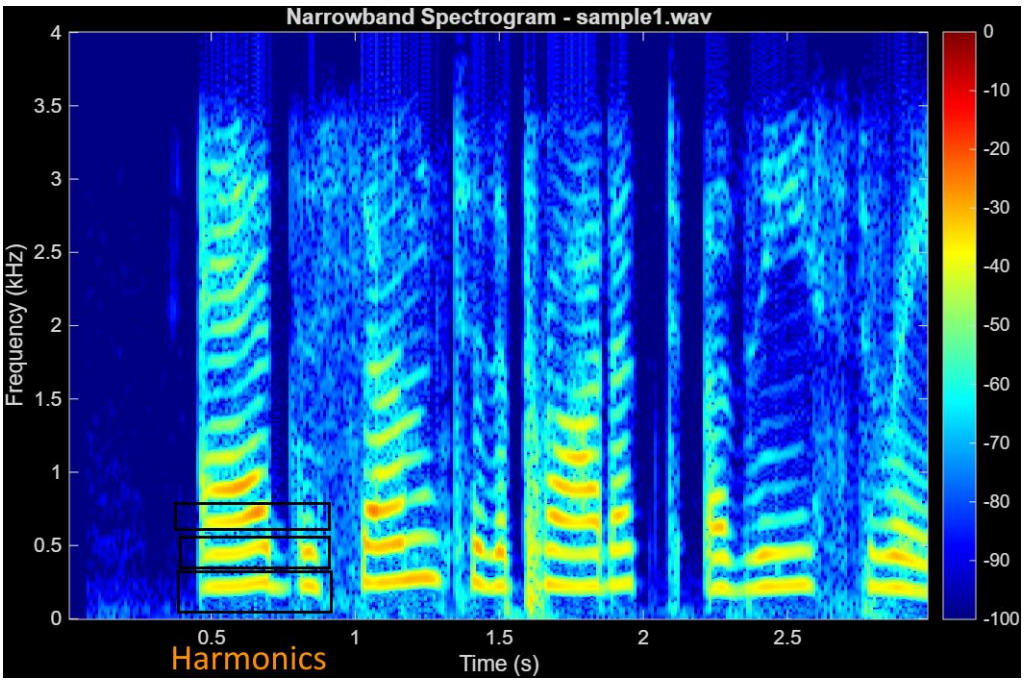
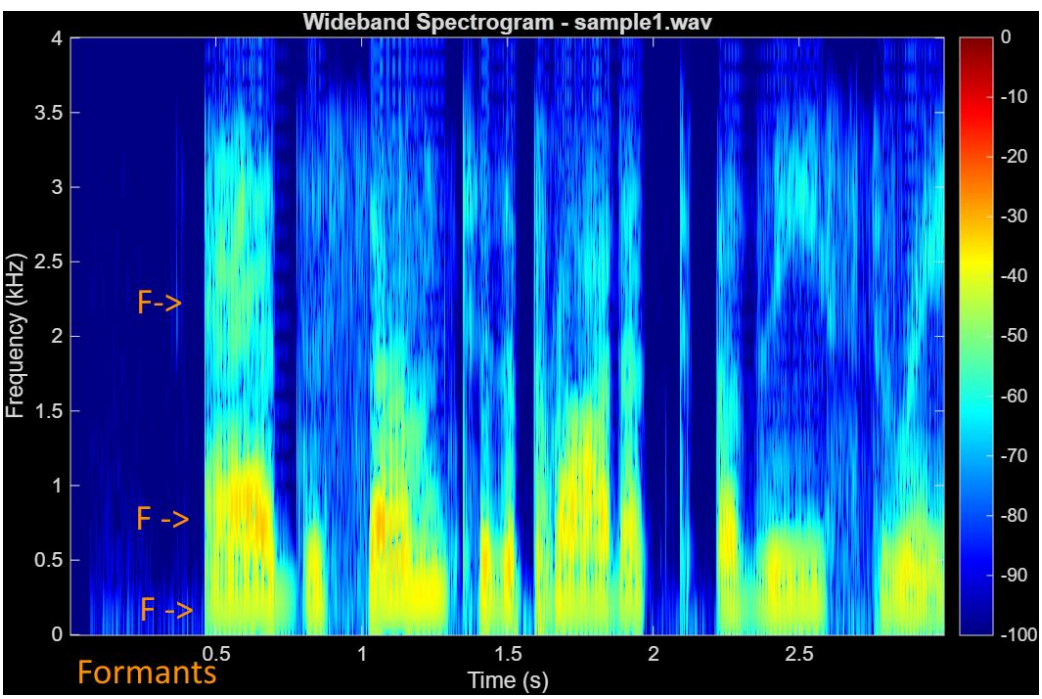
Question 1:


Narrowband Spectrogram - sample1.wav
Frequency (kHz)
Time (s)
Harmonics


Wideband Spectrogram - sample1.wav
Frequency (kHz)
Time (s)
F->
F ->
F ->
Formants

observations:

we can identify the word “add” in the sample 1 in the narrow-band and wide-band at around 0.5 to .75 seconds and the vowel in the word appears in the initial segment of the narrow-band. In this region, the **harmonics** are visible as roughly parallel, wavy yellow and red bands, also notice that the next harmonics are multiple of the first one.

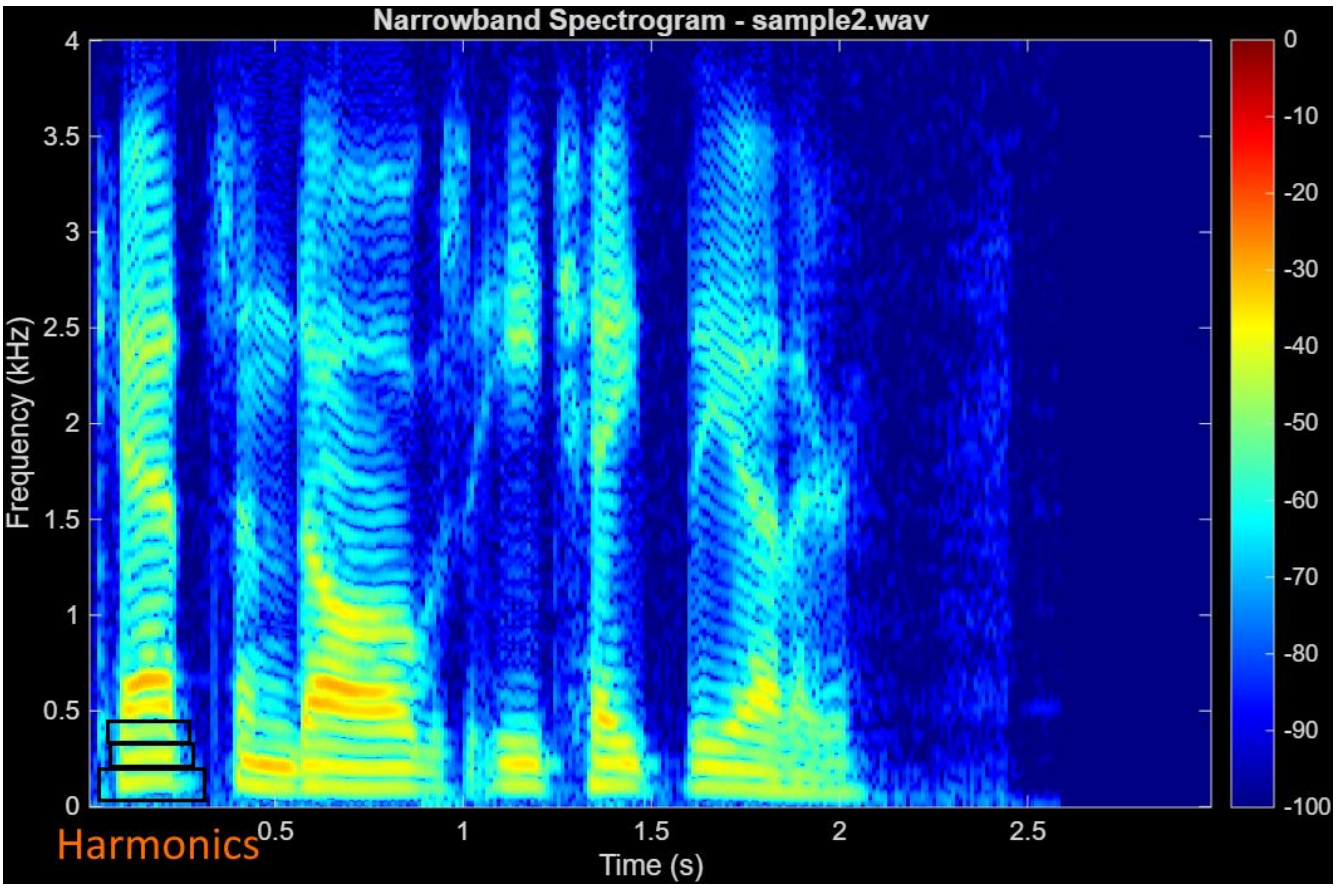
The **fundamental frequency** (F0) is the vertical spacing between the consecutive harmonics.
By measuring the distance between the clear consecutive bands (e.g. the lowest at ~.25 kHz and the next at ~.5kHz, and so on), the approximate spacing is .250 kHz.
Thus, the approximate fundamental frequency is around 250 Hz.

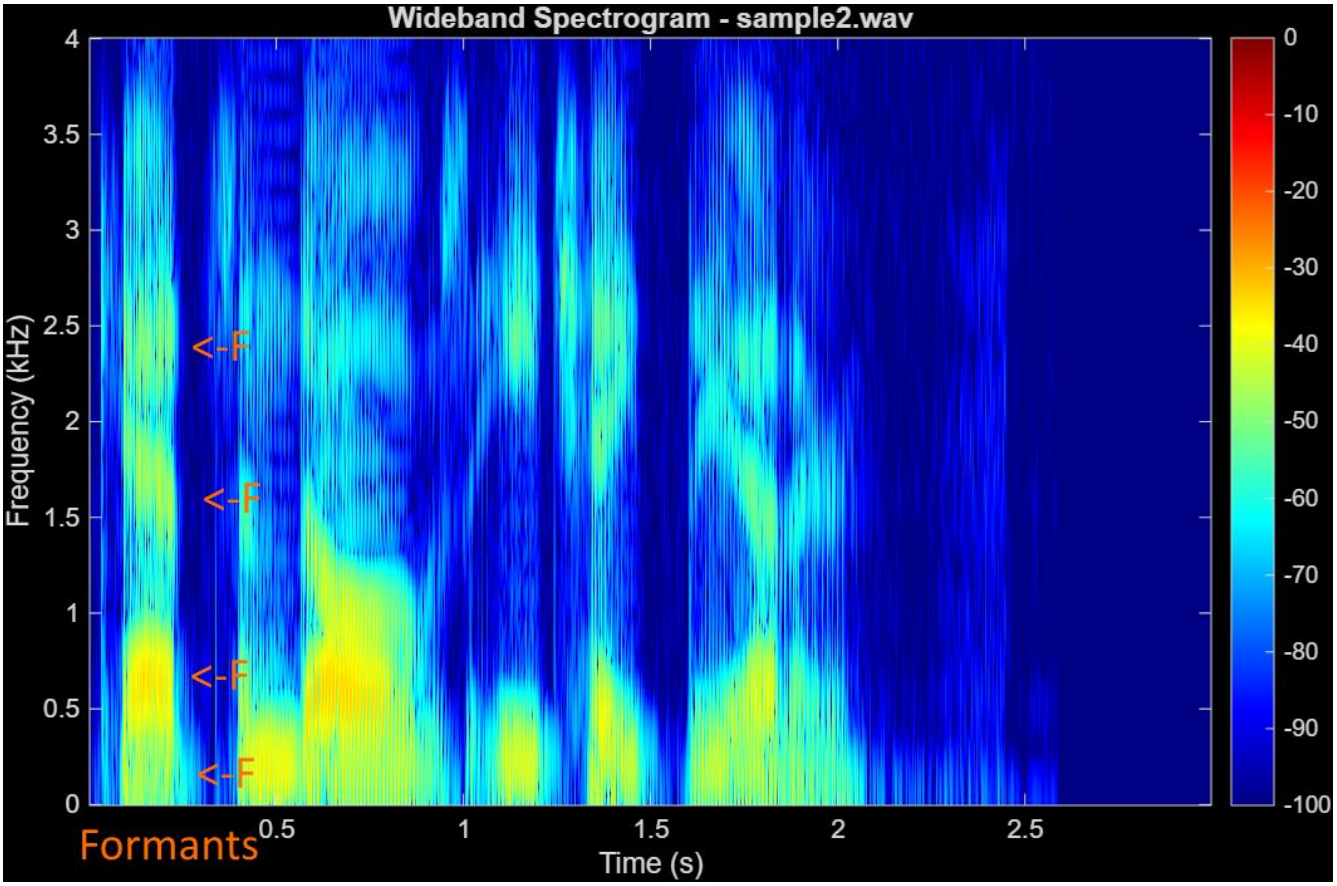
We can also see better the **formant frequency** in the wide-band spectrogram because of the shorts windows.

If we apply the same logic in the sample 2 to find the word “cat” we can see that it appears at .1 to .25 seconds approximately in the narrow-band, again the harmonics are showed in the black boxes in the spectrogram and the F0 is around 160 Hz.

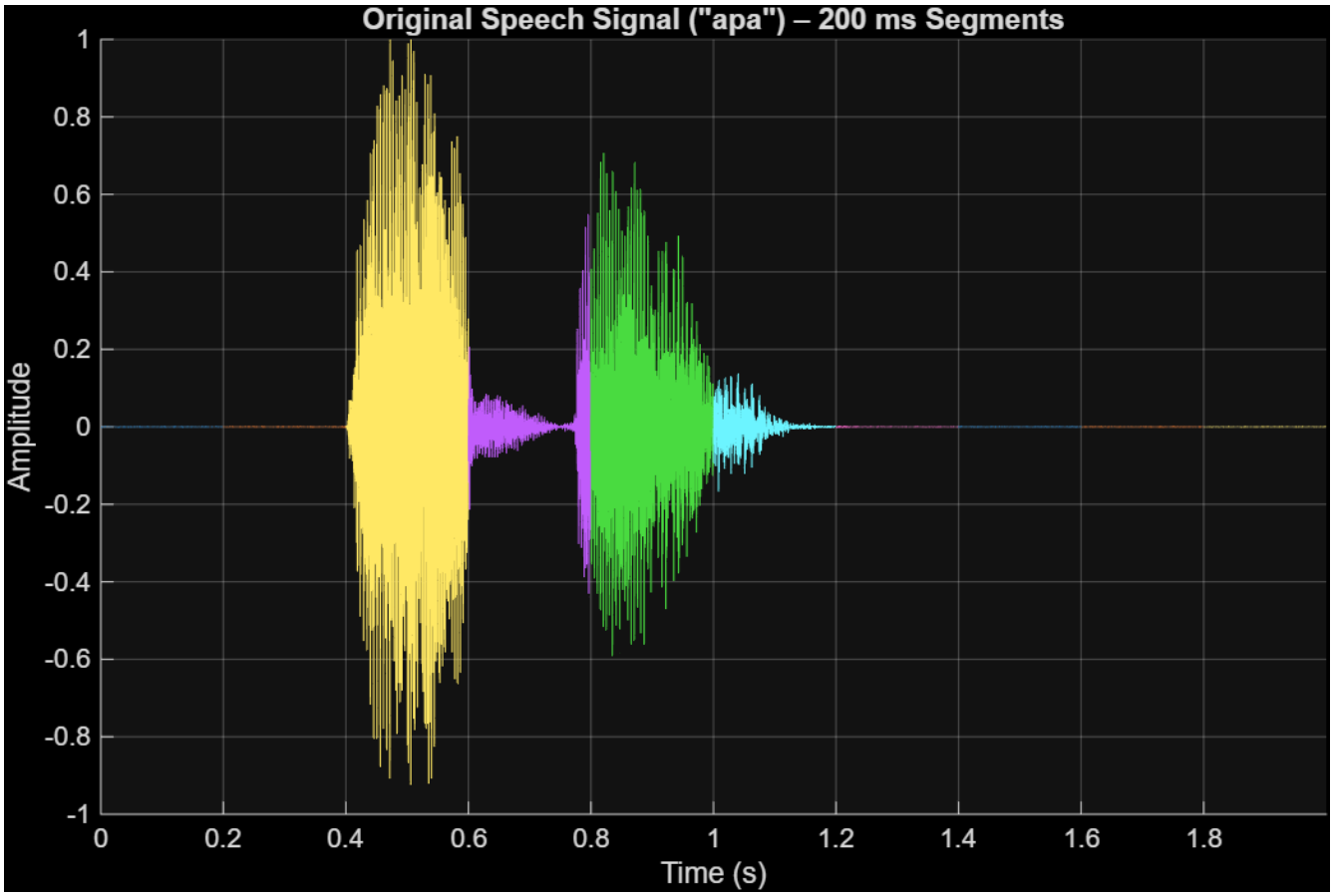
Speaker identification:

The sample 2 has lower F0, so lower pitch as well. Also the harmonics are more densely packed than sample 1 and the formants are lower in frequency meaning that the sound that is produced in the vocal chords has lower pitch.
Based on that information we can clearly say that the sample 1 must be a female and the sample 2 a male.


Narrowband Spectrogram - sample2.wav
Frequency (kHz)
Time (s)
Harmonics

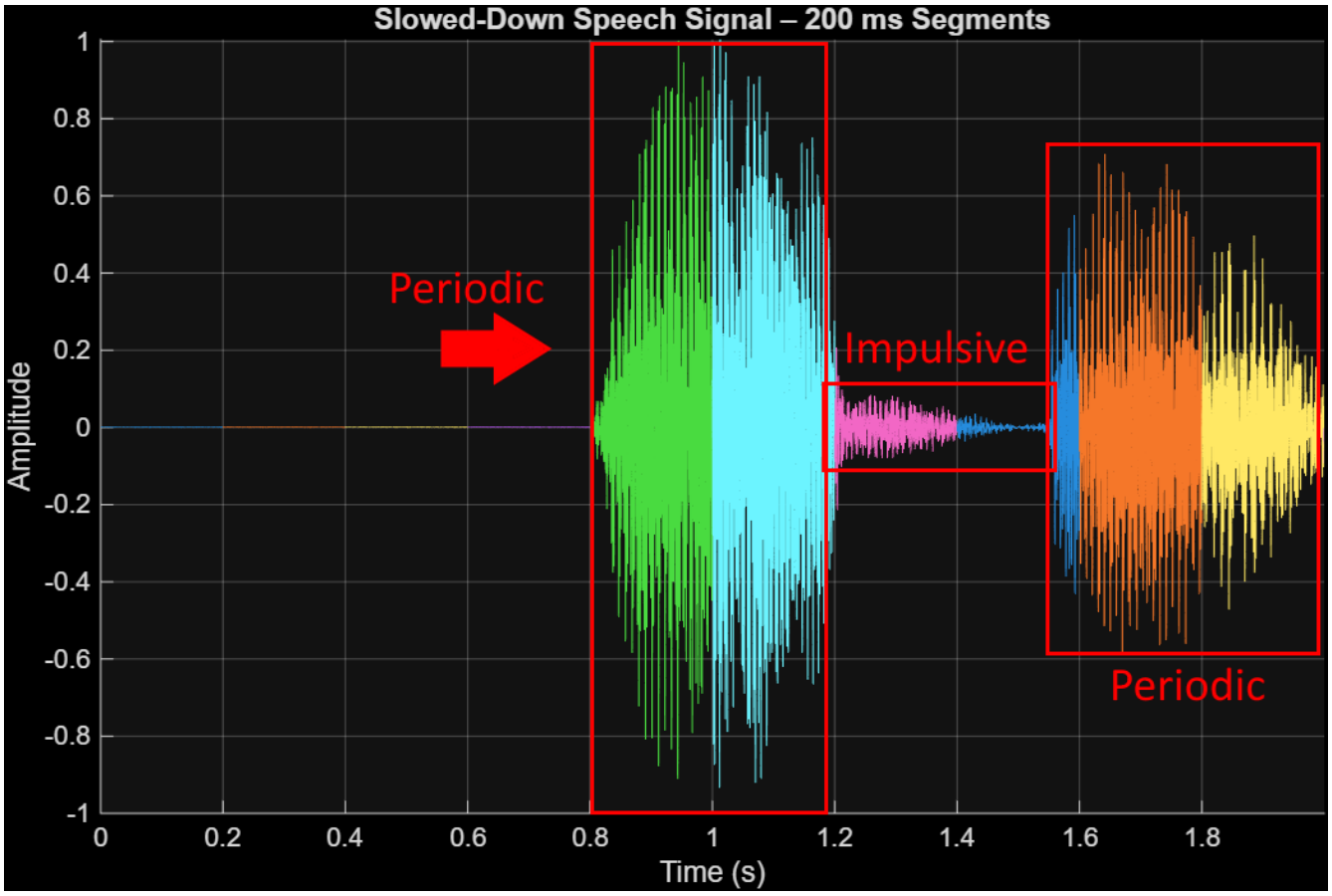

Wideband Spectrogram - sample2.wav
Frequency (kHz)
Time (s)
<-F
<-F
<-F
<-F
Formants

Question 2:



observations:

The **vowel parts** (“a”) have smoother, periodic oscillations (periodic vibration of vocal folds).
The **consonant** (“p”) appears as a short, low-energy segment


Slowed-Down Speech Signal – 200 ms Segments
Amplitude
Time (s)
Periodic
Impulsive
Periodic
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8